

PARTHA GHOSH

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WORK EXPERIENCE

REVE Systems  Jan 2024 – Present
Machine Learning Engineer

- Worked on RAG based chatbot for [REVE CHAT](#) using **GraphRag**, **LightRag**, **Multi-Modal Rag**, **Agentic Rag** and **LangChain**, leveraging hybrid search (BM25 + dense) for enhanced AI chatbot responses.
- Fine-tuned **Wav2vec 2.0** and **Whisper** model for **Bengali STT (কথা)**, also utilizing **PEFT-LoRA**. Analyzed performance on various test sets, including augmented data, and compared models to identify weaknesses.
- Fine-tuned vits model for **Bengali TTS (উচ্চারণ)**
- Developed voice-based agent using an **STT-Chatbot-TTS** pipeline to automate customer interactions and provide seamless conversational services.
- Worked on model compression using **Quantization** and **ONNX** for faster loading, inference times and also for offline services
- Implemented face liveness detection for NTMC biometric authentication project using **OpenCV**, **CNN** and **MediaPipe** for anti-spoofing
- Developed **Time Series Models** and forecasting pipeline for production quantity and sales amount forecasting.
- Developed an chatbot using **BERT** and **Bi-LSTM** with better accuracy
- Worked on the frontend of the dashboard for [REVE CHAT](#), improving usability and interface responsiveness
- **Pytorch | Tensorflow | OpenCV | Langchain | LLM | RAG | React.js**

Codelab FZC  Jun 2023 – Jan 2024
Software Engineer

- Developed a 2D object detection game and a typing game, where players type sentences within a time limit, using Unity for Android and iOS.
- Contributed to developing an admin panel for a project, focusing on implementing CRUD operations in Laravel
- **Unity | C# | Lavavel**

RESEARCH EXPERIENCE

Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla:

We created a 200-hour Bangladeshi Bangla spontaneous speech corpus and fine-tuned the multilingual wav2vec 2.0 model, comparing it with a reading corpus (SUBAK.KO) to enhance ASR performance.

BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark Toward Robust Performance in Challenging Conditions:

Assessed a Bangla ASR model on BanSpeech, a 7.2-hour benchmark; it struggled with spontaneous speech and high out-of-vocabulary rates, highlighting the importance of domain selection.

- Paper Link: ieeexplore.ieee.org/document/10453554

PROJECTS

SUST Student Management System 

Technologies: Django | Python | HTML | CSS | Bootstrap | MYSQL

Managed student, faculty data, departments, courses, and faculty assignments; faculty recorded student marks.

Automatic Attendance System Using Face Recognition 

Technologies: Python | OpenCV | Streamlit

We used face recognition via a Streamlit web app to mark attendance, collecting five pictures per classmate for identification.

Shoesly an E-commerce Website 

Technologies: Node.js | React.js | MongoDB | Express.js

In our app, users, suppliers, admins, and banks have distinct routes for filtering products, confirming orders, and managing transactions between suppliers and customers.

EDUCATION

Shahjalal University of Science & Technology

BSc in Computer Science & Engineering (Graduated)

 Feb 2018 – Mar 2023

CGPA: 3.35/4.00

SKILLS

- **Languages:** Python, Java, C, C++, PHP, C#
- **Machine Learning Technologies:** Pytorch, Tensorflow, OpenCV, LLM, LlamaIndex, Langchain, LangGraph, MCP, n8n, Scikit-learn, MLflow, CI/CD
- **Web Technologies:** Django, FastAPI, JavaScript (React.js, Node.js, Vue.js), HTML/CSS, Laravel
- **Database:** MySQL, MongoDB, Qdrant (VectorDB), Memgraph (GraphDB), Firebase
- **Miscellaneous:** Linux, Shell (Bash), LaTeX, Microsoft Office, Docker, Git
- **Problem Solving:** Solved 600+ problems in different online judge including Codeforces ([partha361](#)), Leetcode ([ppaartha](#)), lightoj ([parthaghoshu](#)), Vjudge ([2017331039](#)), Stopstalk ([ppaartha](#))

ACHIEVEMENTS

1st Runner up in AI for Bangla 2.0 2023

EXTRA-CURRICULAR ACTIVITIES

CSE Society, SUST

General Secretary, Assistant General Secretary, Executive Member

 2019 – 2023

SUST ACM Student Chapter

Joint Secretary

 2021 – 2023

REFERENCE

Dr. Mohammad Shahidur Rahman
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